## FOCUSED PROBLEM

The world's population is rapidly ageing, projected to triple to 1.5 billion people aged 65 years or older by 2050. This demographic shift has resulted in a growing demand for home care services for elderly individuals, particularly those living with chronic illnesses or disabilities.

Chronic diseases are prevalent among older adults and pose various challenges to healthcare systems worldwide. Firstly, treating chronic diseases is expensive and puts a strain on healthcare budgets, particularly in countries with ageing populations. Secondly, there is an anticipated shortage of healthcare workers that may result in inadequate care for the growing number of older adults with chronic diseases. Thirdly, preventive care is crucial in preventing or delaying the onset of chronic diseases, and providing preventive care to older adults is essential to ensure that they remain healthy and independent for as long as possible.

Obtaining real-time monitoring of health data is our main focus since it is very important to support preventive care and precision medicine. For example, one of the most common chronic illnesses affecting the elderly which is Parkinson's disease, a chronic neurodegenerative condition that affects muscle control and the symptoms become worse over time. Globally, it is estimated that over 10 million people suffer from Parkinson's disease. This number is expected to rise as the population ages. In Taiwan the death rate of parkinson diseases had been increasing more than 10% each year and in 2017 the number almost reached 600. The disease is more common among the elderly, generally affecting people over the age of 60, although younger-onset cases are not rare.

Some of the common symptoms include tremors (shaking in the hands, arms, legs or face), stiffness or rigidity of the muscles, slowed movement (bradykinesia), poor balance and coordination, etc. These are generally staged using the Hoehn and Yahr scale which ranges from stage one to five, where stage one is mild symptoms, usually on one side of the body, to stage five where the patient is bedridden or wheelchair-bound and need full-time assistance. With continuous real-time monitoring of these conditions, we aim to be able to do fall alarm by observing patients' movements and activities since Parkinson's disease patients tend to fall often while it might occur further complications if it's not being treated immediately.

To maintain the health and well-being of individuals with Parkinson's disease, providing specialised home care services is crucial, especially as the falling prevention services. However, the cost of such care can be prohibitively expensive for many households, especially those with

low incomes. Therefore, developing affordable and accessible home care services that cater to the needs of elderly individuals with Parkinson's disease and other chronic illnesses or disabilities is critical.







According to 17 sustainable development goals of the United Nations, the monitoring system for Parkinson's patients plays a crucial role in promoting good health and well-being (SDG 3) while aligning with responsible consumption, production, and sustainable development principles (SDG 12). This "Wi-Care" device offers a lifeline to individuals with Parkinson's disease by providing real-time data monitoring on their symptoms and gait captures, then provide valuable insights for effectiveness assessment and falling detection. Doing so contributes to better quality of life, increased independence, and reduced healthcare costs. Additionally, the development and use of our device emphasize responsible consumption and production practices as they reduce the wastefulness often associated with inefficient treatments and hospitalizations. In a broader context, the sustainable development goal of ensuring good health and well-being for all is intertwined with the responsible use of resources, making advancing Parkinson's monitoring technology a significant step toward achieving a more sustainable and equitable healthcare system.

**PROPOSED SOLUTION: 'Wi-Care'**

"Wi-Care", is a smart home caregiver system based on Wifi sensing technology that is specifically designed for Parkinson's patients. This system is ultized to address contactless health monitoring in elderly individuals, specifically to support the home-based long-term care provider while providing a range of other smart features to support the elders' health and well-being. The system utilises wifi sensing technology to detect respiratory symptoms and monitor vital signs such as respiratory rate, heartbeat rate, and sleep tracking data in real-time, allowing for early detection and intervention.



**MAIN FEATURES**

a. **Fall Detection & Activity Recognition:**

WiFi sensing technology uses wireless signals emitted by WiFi access points to differentiate between various types of motion, including walking, sitting, standing, and falling, with an accuracy of up to 94%. By analysing disturbances in the wireless signals, WiFi sensing can detect falls and send immediate alerts to caregivers or emergency services, allowing for prompt assistance.

Furthermore, WiFi sensing can also monitor and track older adults' daily activities, providing insights into their health and well-being. For instance, this technology can track the number of steps taken, time spent sitting, and movement frequency.

b. **Remote Monitoring:**

Caregivers can access a secure mobile app to view real-time data from the smart home system, including the patient's activity, movement patterns, and environmental conditions.The system sends alerts to caregivers' devices when predefined thresholds or unusual events occur. For example, if the patient is inactive for an extended period or if a fall is detected, the caregiver will receive an immediate notification. Caregivers can also access historical data, allowing them to track trends in the patient's behavior and health over time. This information can be valuable for making informed decisions about the patient's care and treatment plan adjustments. Remote monitoring provides caregivers with peace of mind, knowing that they can check on the patient's well-being and respond quickly to any issues, even when they are not physically present.

**c. Medication Reminders:**

The system allows caregivers to input the patient's medication schedule, specifying medication names, dosages, and times. This schedule can be tailored to the patient's prescription plan. The second is automatic notifications: When it's time for a dose, the system sends notifications or alarms to both the patient and the caregiver's devices. These reminders can be in the form of text messages, app notifications, or even audible alerts. Our system can also provide medication logging by enable patients to confirm that they've taken their medications. This confirmation can be logged and shared with caregivers, providing peace of mind. If a patient forgets to take their medication within a certain timeframe, the system can send alerts to the caregiver, who can then follow up to ensure the patient's safety and adherence to the treatment plan.

**d. Gait Analysis:**

The system uses WiFi signals and sensors strategically placed around the home to monitor the patient's movements, including walking patterns. Initially, the system establishes a baseline for the patient's gait and mobility. It learns what is normal for the patient. Over time, the system continually monitors the patient's gait, assessing parameters like step length, walking speed, and stability. If the system detects significant deviations from the established baseline, it can send alerts to caregivers. For example, if the patient's walking speed slows down or if they start exhibiting a shuffling gait, it may signal a potential issue. The most important is the early intervention. The early detection of changes in gait can allow for timely intervention, adjustments in treatment plans, or physical therapy to address mobility issues associated with Parkinson's disease.

**e. Voice-Activated Assistance:**

The system connects with voice-controlled devices like Amazon Alexa or Google Assistant, allowing the patient to control various aspects of their home environment. Patients can use voice commands to turn lights on or off, adjust the thermostat, control entertainment systems, and more. This reduces the physical demands on the patient, which can be beneficial if they have mobility issues. Patients can customize voice commands to suit their preferences and needs. For example, they can create a command to set the lighting to a specific level for reading. Voice-activated systems can also be used to call for help in emergencies. For instance, a patient can use

a predefined voice command to request assistance or report a fall.

**What is WiFi Sensing?**

WiFi sensing technology operates by analysing Channel State Information (CSI), which refers to the wireless communication details between a sender and receiver. By analysing the level of signal attenuation caused by various factors, such as movement or obstruction, the CSI enables the detection of changes in the wireless signal, which can provide valuable insights about the surrounding environment or objects.

The popularity of sensing and monitoring systems in the field of human healthcare has grown significantly. This trend can be attributed to the knowledge-based economy and the notable advancements in our longevity and living standards. These systems offer the ability for individuals to continuously detect daily activities and fluctuations in vital signs, all within the confines of their homes. By facilitating simple, long-term, and uninterrupted health monitoring in our everyday living environment, it becomes feasible to identify signs of illness and physiological decline that may go unnoticed during brief clinical consultations. Additionally, these monitoring systems can be integrated with deep learning techniques to track physical behaviour and vital signs, enabling the identification of potential health issues and physiological changes. The collected information can be seamlessly incorporated into smart homes to support our daily lives.

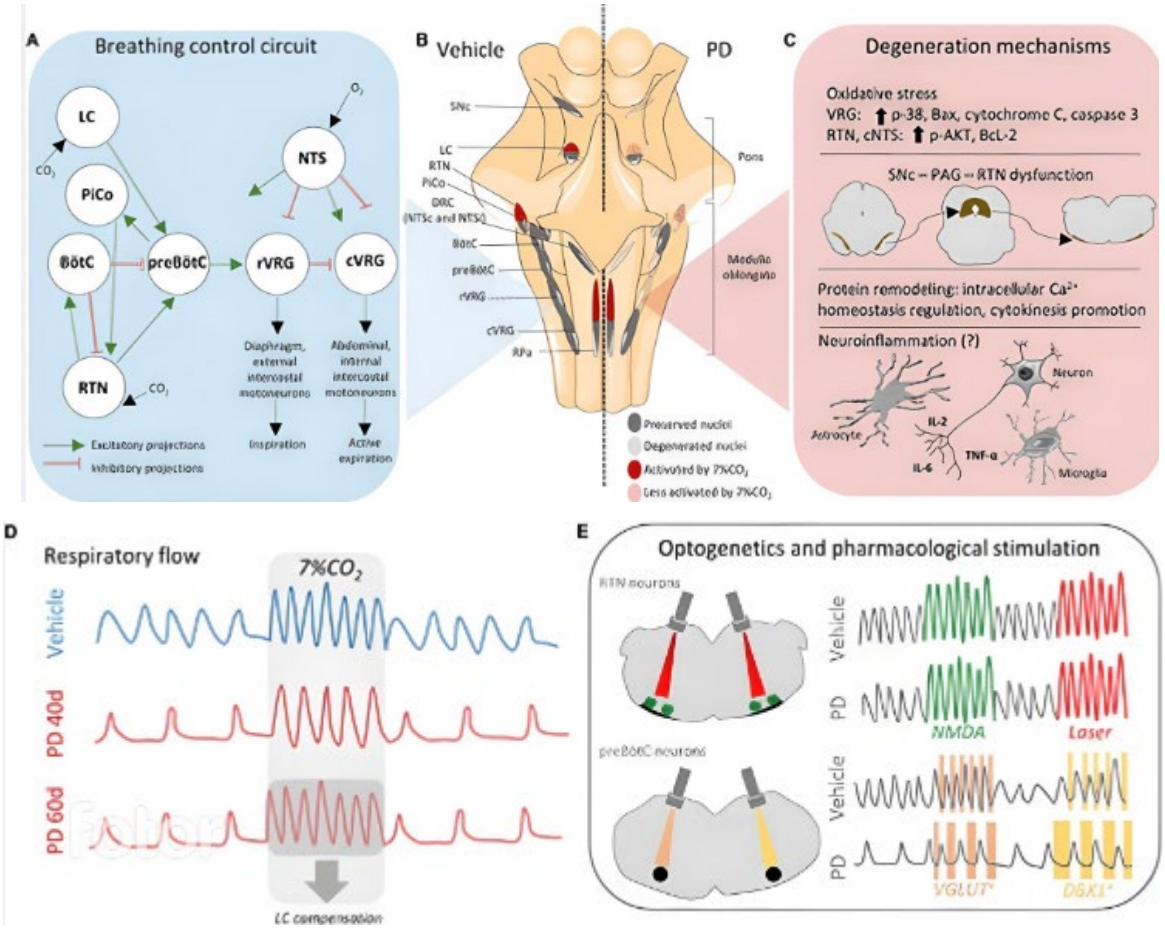
**Augmented Analytics for Long-Term Care (LTC)**

Augmented analytics identifies patients at risk of health issues from medical records, surveys, and Wi-Fi sensing. Early detection enables timely interventions, preventing complications. It also facilitates continuous health monitoring, enabling personalised care adjustments for better outcomes and well-being. Real-time monitoring is essential for long-term care of older adults, as it can help detect changes in biological patterns and identify potential health problems.

For example, Parkinson's disease is a brain disorder that causes unintended or uncontrollable movements, such as shaking, stiffness, and difficulty with balance and coordination. Symptoms usually begin gradually and worsen over time. As the disease progresses, people may have trouble walking and talking. Parkinson's has four primary symptoms: tremors in hands, arms, legs, jaw, or head, muscle stiffness where muscle remains contracted for a long time,

slow movement, and impaired balance and coordination, sometimes leading to falls. In addition, Parkinson's disease often develops a Parkinsonian gait that tends to lean forward, take small, quick steps, and reduce swinging of their arms. They also may have trouble initiating or continuing movement. However, despite significant lifestyle and mental changes, PD patients tend to fall easily because of the gait's uncontrollable movements. Falls in older adults are common, but their side effects might cause serious problems, like fractures.

Therefore, fall prevention in the elderly, especially PD patients, is essential. In this study, we will use Wi-Fi sensing technology to assist medical professionals in tracking the effectiveness of the treatments given to patients. Besides that, we will also add an alarm system through the app to notify when the patient is about to fall or has fallen for prevention and alert people nearby to help. The Wi-Fi sensing technology is based on gestures and movement detection, combined with height measurement to classify different activities and falling detection.



## DATA PROCESSING METHODOLOGY

1. Raw data collection: Data is collected from various sources, including WiFi sensor readings, motion data, and user inputs.
2. Data analysis: The collected data is analysed using the time-splitting method to categorise it into indicators and targets. This method allows for scalability, parallel processing, granularity, resource management, incremental processing, and data exploration.
3. Machine learning: The data is recorded in a model database (CSV file) for use as input data for machine learning. Machine learning is leveraged to develop anomaly detection and improve prediction accuracy using historical data.
4. Visualisation and Insight generation: The predictions generated by AI are presented as insights for app users through data visualisation, such as graphs and tables. This allows users to easily monitor the obtained data, such as average heartbeat, active rates derived from motion data, real-time, and predicted percentage of getting diseases such as COPD and asthma.
5. User action: Calculated functions and insights are derived, culminating in user actions. The project generates graphs and average data that users can access through their application accounts.

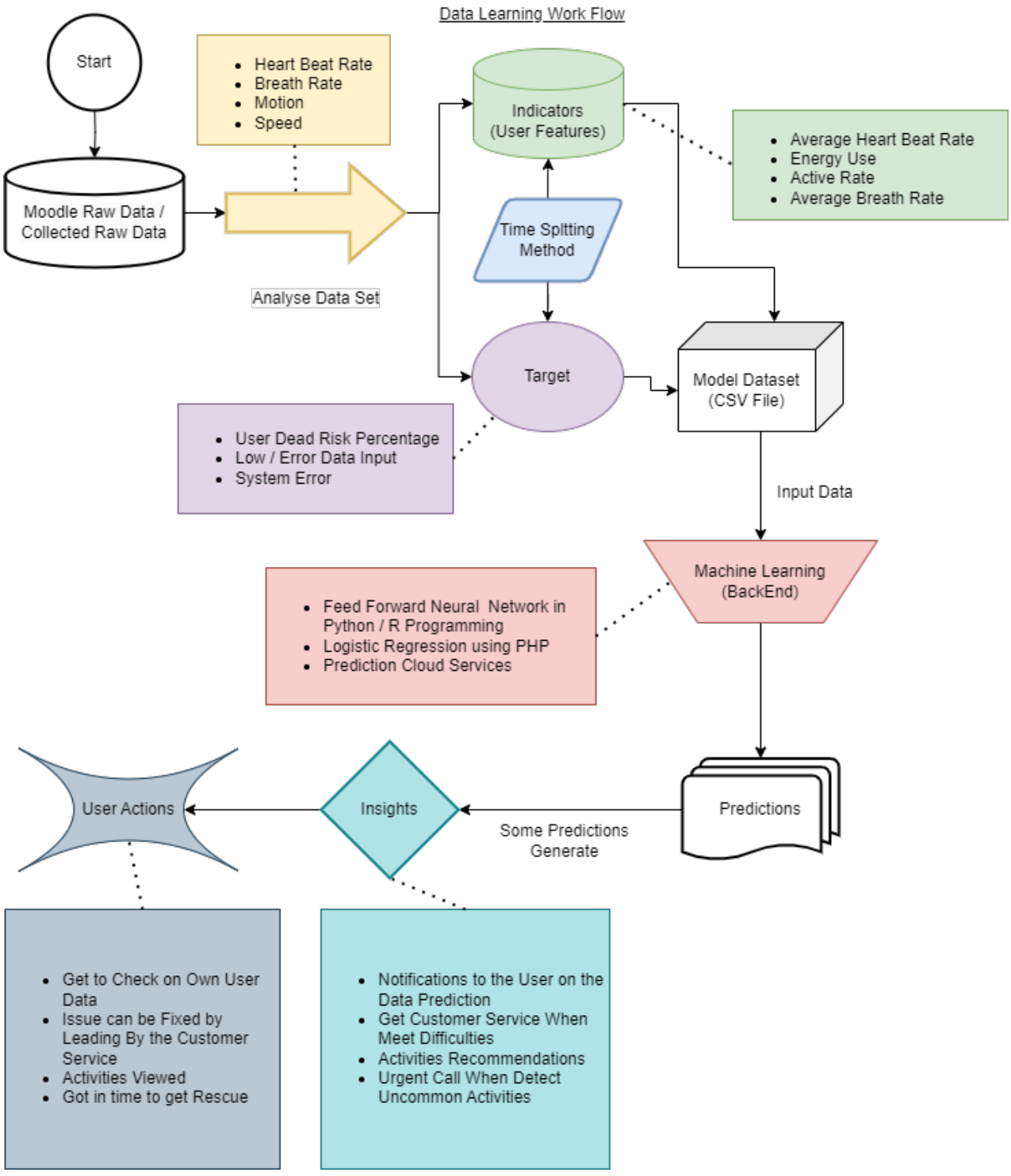



Figure 4: Data Processing Methodology

**THE USE OF MOBILE APP**

- The app allows users to connect their phone to the WiFi sensing device, which can provide input for a range of data that the device itself cannot obtain such as past health records and medical information.
- Visualise data and make it more accessible and user-friendly by displaying the various data in a clear and easy-to-understand format, such as graphs or charts.
- Real-time monitoring and remote access for caregivers to monitor users' health data, breathing patterns, activity status, etc.
- The app can also provide notifications or alerts if there are any significant changes in respiratory data.
- Emergency notifications will be sent to the user's caregivers or family via the app whenever fall detection or onset of respiratory symptoms happens.

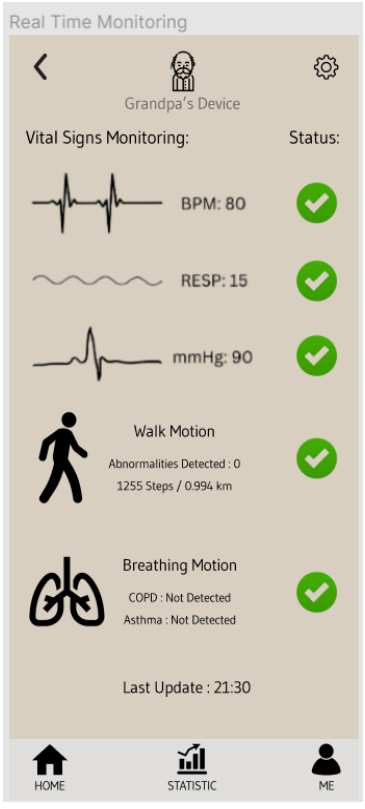
## APP PROTOTYPE

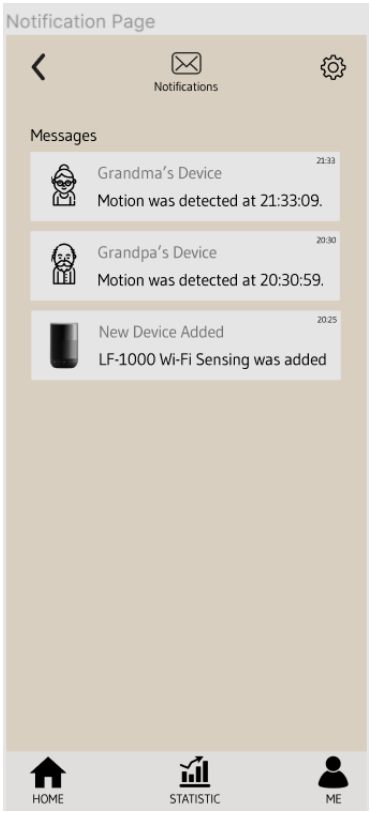


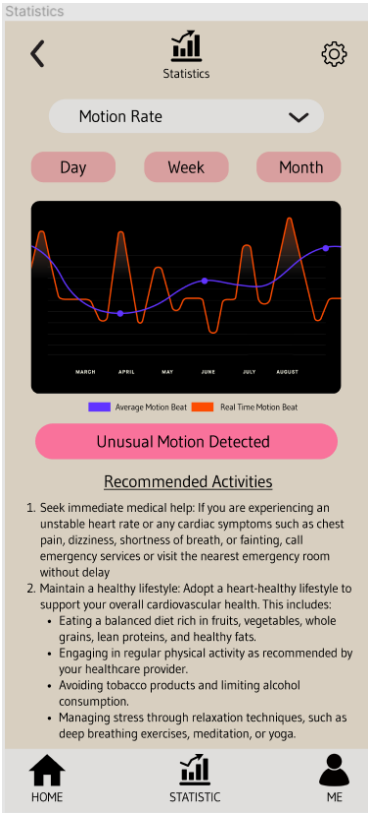






## COMPETITOR ANALYSIS

Table 1. Comparison to other health monitoring technologies

| **Competitors** | **Description** | **Pros** | **Cons** |
|---|---|---|---|
| Wearable Health Monitors | Include devices like smartwatches and fitness trackers that individuals wear on their bodies to track various health metrics such as heart rate, activity levels, and sleep patterns. | •Real-time data<br>•Personalisation with mobile apps | •Intrusive: Some users find wearables uncomfortable to wear 24/7.<br>•Limited battery life<br>•Expensive for higher-end models |
| AI-Powered Health Monitoring System | AI-powered health monitoring apps leverage smartphone sensors to track various health metrics, such as heart rate and sleep patterns. Use machine learning algorithms for data analysis. | •Accessibility: Most people have smartphones.<br>•Continuous monitoring: Apps can provide ongoing health data when used regularly. | •Limited metrics<br>•Inaccuracy<br>•Lack of healthcare provider integration |
| Monitoring cameras (Examples include Arlo & Nest Cam) | Camera technology allows caregivers or family members to remotely monitor the activities of an elderly individual in their home. | •Remotely accessible<br>•Real-time monitoring | •Invasion of privacy<br>•Loss of autonomy for the individual |

## MARKET INSPECTION

'Wi-Care' has potential in the growing home healthcare market, particularly in addressing. According to market research, the global home healthcare market size is projected to reach USD 274.7 billion by 2025 from USD 181.9 billion in 2020, at a CAGR of 8.6%. This growth is driven by factors such as the ageing population, rising prevalence of chronic diseases, and increasing demand for personalised healthcare.

In today's market, there is a growing demand for innovative products that cater to the immediate response needs of the elderly during sudden events that pose a danger to their well-being, such as Parkinson's Disease (PD), falls, heart attacks, asthma attacks, and more. The global PD market is substantial, with an estimated 10 million people living with PD globally making it the second most common neurodegenerative disorder worldwide. This number is projected to increase due to ageing populations. In Taiwan, the prevalence rate of PD has recorded a yearly increase of 7.9%, and there is an increasing trend in all age groups, especially those in their 80s. In Southeast Asia, the PD prevalence rate has skyrocketed with a surprising 174.08% from 1990 to 2019. Thus, continuous monitoring of PD symptoms is getting more essential for timely intervention, but existing solutions have limitations, including discomfort and invasiveness.

WiFi sensing technology is currently supported by certain types of chipsets, and its compatibility with devices is expected to increase in the coming years. According to market research, the number of compatible devices will increase from USD 156.77 million in 2022 to USD 192.67 million in 2023 and accelerate after 802.11bf standardisation in 2024. This suggests that there is growing interest and investment in WiFi sensing technology, which could create opportunities for our product.

**BUSINESS MODEL**

## Key Partners

Healthcare Providers & Retailers
Technology Suppliers, Manufacturers, & Distributors
Government Agencies & Industry Players

## Key Resources

WiFi Sensing Technology & Infrastructure
Funds
Data Storage & Analysis Capabilities
R&D Team
Partnerships & Collaborations
Regulatory Approvals
Brand Reputation & Customer's Loyalty
Customer Support Team

## Customer Segments

Parkinson's Disease Patient
Caregivers
Healthcare Institutions & Professionals
Smart Home Devices Enthusiasts or Early Adopters

## Key Activities

Short Term:

- Product Development
- Market Research
- Supply Chain & Partnerships
- Pre-launch Marketing & Promotion
- Public Relations

Mid Term:

- Expansion in Taiwan & Approach International Market
- Examine User's Usage & Improve Their Experience
- Expand Company Size & Tailor Future Roadmap
- Loyalty Campaign

Long Term:

- Expansion in other countries
- Introduce New Features
- Diversification
- Third Party Integration

## Key Value Propositions

Non-Intrusive Monitoring
Early Detection & Intervention
Real-time Personalized Health Insights
Power Saving & Environmental-friendly
User-friendly, Easy Installation & Affordable

## Customer Relationships

Continuous Customer Support
Regular Software Updates & Improvements
Survey & Feedback Gathering
After-sales Support
Warranty
Loyalty Campaign

## Channels

Website, App & e-commerce platforms
Partnerships with healthcare institutions for distribution
Collaboration with smart home device manufacturers for bundled offerings
Healthcare seminars & conferences
Social media
Retail Partners

**COST PER UNIT**

| **Cost Per Unit** | |
|---|---|
| **Material Used** | **USD** |
| **ESP32 CPU** | **12** |
| **3D Printing Cover (Recycled Material)** | **25** |
| **Motion Sensors** | **5** |
| **STM32 Ultra Low Power MCU** | **3** |
| **TOTAL** | **45** |

**COST STRUCTURE: 3 YEARS' FINANCIAL PLAN**

| **Financial Year 2024** | | | |
|---|---|---|---|
| | Project | Content | Cost Estimate (USD) |
| **1** | **Early Product Development** | **Wifi Sensor Design, AI System Establishment AND Wifi Sensor Settings with the Development Board** | **$2,000** |
| | | **Wifi Sensor Testing AND Prototype Design** | **$3,000** |
| **2** | **Expert Consulting** | **Business Model, Regulations and Patent Experts** | **$7,000** |

| | | | |
|---|---|---|---|
| **3** | **Patent Layout** | **PA AND PCT** | **$15,000** |
| | | **Invention Patent, Utility Model Patent, Design Patent** | **$900** |
| **4** | **Salary** | **Total Salary For 2023** | **$12,000** |
| **5** | **Operation in Taiwan** | **Business Registration** | **$38** |
| | | **Accounting and Audit (Every 14 Months)** | **$1,400** |
| | | **Laboratory and Office Rental (Per Year)** | **$25,000** |
| | | **Labour and Health Insurance (Per Year)** | **$1000** |
| | | **Internet (HiNet) (Per Year)** | **$600** |
| | | **Utility Bill (Per Year)** | **$1,200** |
| **6** | **Regulatory Certification** | **(Taiwan) BSMI + NCC: certificate (per year)** | **$800** |
| **7** | **Others** | **Consumables and Transportation** | **$1,500** |
| | | **TOTAL** | **$ 69,138**<br>**NT$ 2,166,079**<br>**.71** |

| **Financial Year 2025** | | | |
|---|---|---|---|
| | Project | Content | Cost Estimate |

| | | | (USD) |
|---|---|---|---|
| **1** | **Product Development** | **Wifi Sensor Alpha & Beta Testing** | **$5,000** |
| | | **Application Development AND Design** | **$1,000** |
| | | **Adjustable Element Design AND Mass Production Test** | **$5,000** |
| **2** | **Expert Consulting** | **Wifi Technology AND Device Testing Experts** | **$3,000** |
| **3** | **Salary** | **Total Salary For 2024** | **$60,000** |
| **4** | **Operation in Taiwan** | **Accounting and Audit (Every 14 Months)** | **$1,400** |
| | | **Laboratory and Office Rental (Per Year)** | **$25,000** |
| | | **Labour and Health Insurance (Per Year)** | **$1000** |
| | | **Internet (HiNet) (Per Year)** | **$600** |
| | | **Utility Bill (Per Year)** | **$1,200** |
| **5** | **Regulatory Certification** | **(Taiwan) BSMI + NCC: certificate (per year)** | **$800** |
| **6** | **Others** | **Consumables and Transportation** | **$3,500** |
| | | **TOTAL** | **$ 107,500**<br>**NT$ 3,367,953.50** |

| Financial Year 2026 | | | |
|---|---|---|---|
| | **Project** | **Content** | **Cost Estimate (USD)** |
| **1** | **Product Development** | **Mass Production** | **$20,000** |
| | | **Research AND Develop New Functions** | **$5,000** |
| **2** | **Expert Consulting** | **Marketing Experts** | **$1,000** |
| **4** | **Salary** | **Total Salary For 2025** | **$90,000** |
| **5** | **Operation in Taiwan** | **Business Registration** | **$38** |
| | | **Accounting and Audit (Every 14 Months)** | **$1,400** |
| | | **Laboratory and Office Rental (Per Year)** | **$25,000** |
| | | **Labour and Health Insurance (Per Year)** | **$1000** |
| | | **Internet (HiNet) (Per Year)** | **$600** |
| | | **Utility Bill (Per Year)** | **$1,200** |
| **6** | **Regulatory Certification** | **(Taiwan) BSMI + NCC: Laboratory test + certificate (per year), (Singapore) IMDA: Laboratory test +** | **$900** |

| | | | |
|---|---|---|---|
| | | **certificate (per year)** | |
| **7** | **Others** | **Consumables and Transportation** | **$3,500** |
| | | **TOTAL** | **$ 146,138**<br>**NT$ 4,578,474.31** |

**As Taiwan is Singapore's Mutual Recognition Agreement (MRA) partner, the laboratory test fee may be waived upon doing so in Taiwan. Thus, we may apply for it and it is subject to approval.*

**REGULATORY**

Adherence to relevant regulations stipulated by the National Communications Committee (NCC) and Bureau of Standards, Metrology and Inspection (BSMI) and Infocomm Media Development Authority (IMDA Singapore, in mid-term) to obtain the certification.

| | |
|---|---|
| **Product Name** | Wi-Care |
| **Equipment Category** | New Equipment Type |
| **Equipment Type** | IoT User Equipment |
| **Documents for Submission (including but not limited to)** | 1) User manual |
| | 2) Specification document |
| | 3) Electric circuit diagram |
| | 4) Inspection report (from the laboratory approved by NCC/BSMI/IMDA) |
| | 5) Business registration certificate |

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